

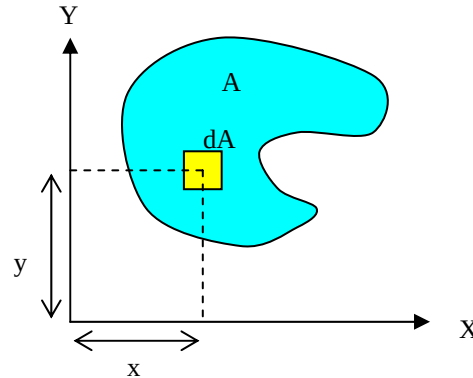
Properties of symmetrical and unsymmetrical sections

Centroid

The coordinate of the centroid of an area A is defined by

$$\bar{x} = \frac{\int_A x dA}{\int_A dA}$$

$$\bar{y} = \frac{\int_A y dA}{\int_A dA}$$

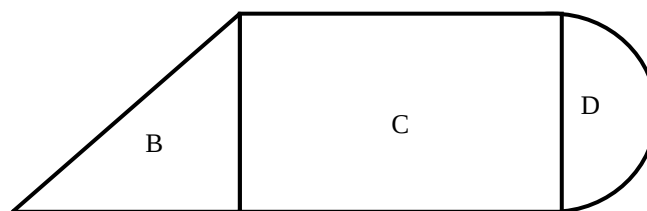


Where x and y are the co-ordinate of the differential element dA . And x and y are distances from any reference line or axis. And $\bar{x}$ and $\bar{y}$ are measured from these reference axes.

Complex area

For more complex cross sections we must divide the section into regular parts/segments. The co-ordinates of the overall centroid can then be expressed in terms of the co-ordinates of the centroid of its parts/segments.

This is better explained by the following example:



$$\bar{x} = \frac{\int_A x dA}{\int_A dA} = \frac{\int_B x dA + \int_C x dA + \int_D x dA}{\int_B dA + \int_C dA + \int_D dA}$$

Where A is the total complex area

But $\int_A x dA = A \cdot \bar{x}$, which results in

$$\bar{x} = \frac{\sum_i x_i A_i}{\sum_i A_i}$$

Similarly

$$\bar{y} = \frac{\sum_i y_i A_i}{\sum_i A_i}$$

Location of the centroid of member with a regular complex section.

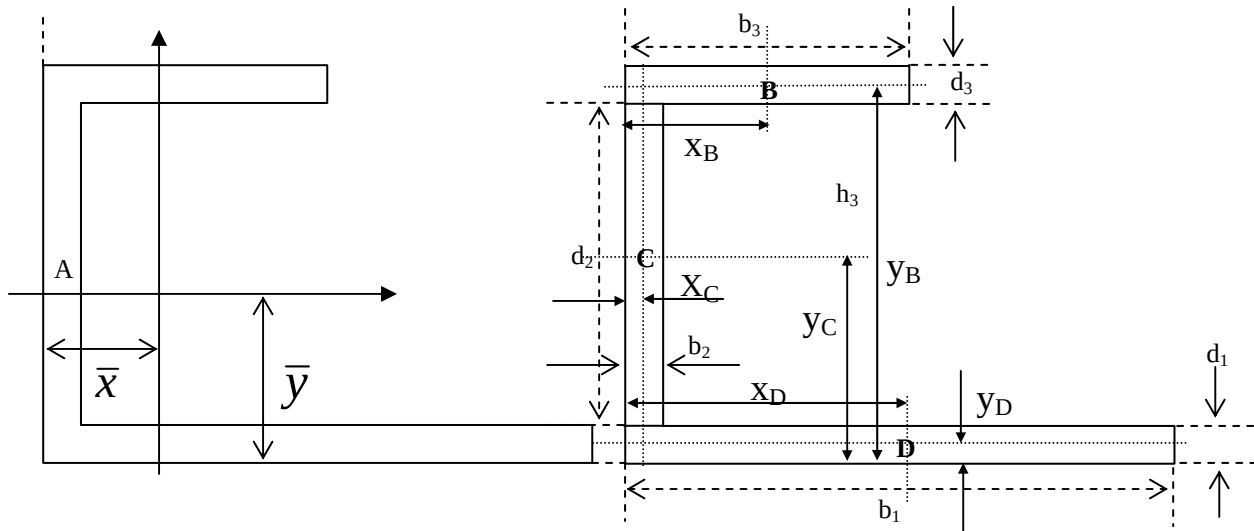
To determine the position of the centroid of a complex section shown below we must;

- Divide the section into regular shaped segments.
- Assume a reference set of axes
- Calculate the area of each segment
- Identify the position of the centroid of each regular segment.
- Find the distances X_i and Y_i between the centroid of each segment and the reference axes.
- The $\bar{X}$ or $\bar{Y}$ co-ordinates of the centroid of any complex section is then defined as,

$$\bar{X} = \frac{\sum_1^n (area)_i \times X_i}{\sum_1^n (area)_i}$$

$$\bar{Y} = \frac{\sum_1^n (area)_i \times Y_i}{\sum_1^n (area)_i}$$

This is better explained by the following example:



$$\bar{x} = \frac{\int_A x dA}{\int_A dA} = \frac{\int_B x_B dA + \int_C x_C dA + \int_D x_D dA}{\int_B dA + \int_C dA + \int_D dA}$$

Where $dA = x_i \times y_i$ and A is the total complex area and $i = B, C$ or D

But $\int_A x dA = A \cdot \bar{x}$, which results in

$$\bar{x} = \frac{\sum_i x_i A_i}{\sum_i A_i}$$

Similarly about Y axis

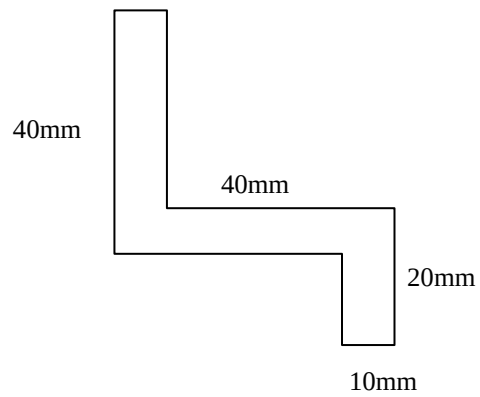
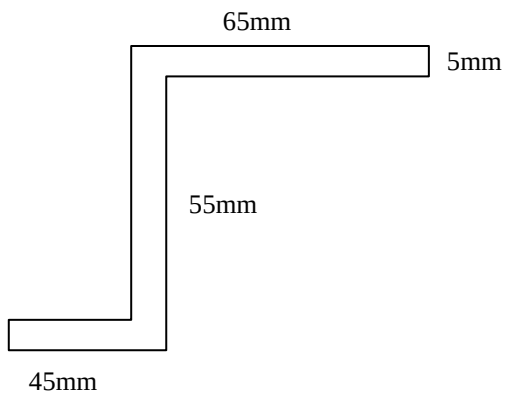
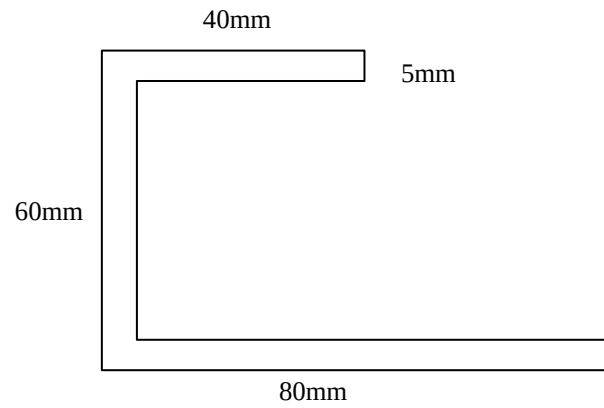
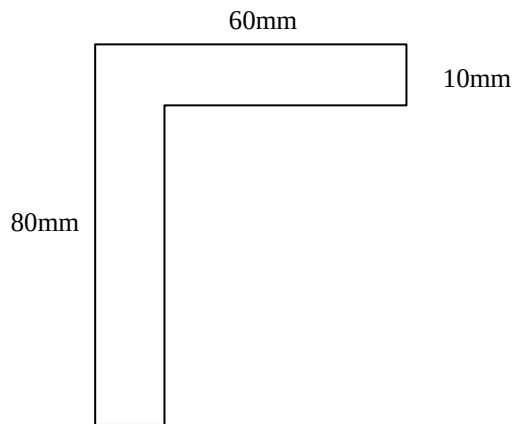
$$\bar{y} = \frac{\int_A y dA}{\int_A dA} = \frac{\int_B y_B dA + \int_C y_C dA + \int_D y_D dA}{\int_B dA + \int_C dA + \int_D dA}$$

But $\int_A y dA = A \cdot \bar{y}$, which results in

$$\bar{y} = \frac{\sum_i y_i A_i}{\sum_i A_i}$$

Example:-

Calculate the position of the centroid of the sections shown below,



Second Moment of Area, I

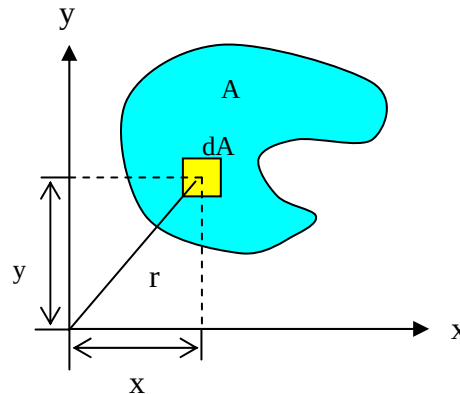
The generic expression for the second moment of area I about x and y axes are;

$$I_x = \int_A y^2 dA$$

$$I_y = \int_A x^2 dA$$

For the product of moment of area

$$I_{xy} = \int_A xy dA$$



Polar second moment of area

$$J_o = \int_A r^2 dA = I_{xx} + I_{yy}$$

Where $r^2 = x^2 + y^2$

For regular shapes, I values are readily available and some of the most commonly used ones are listed below.

For rectangle $I_{xx} = \frac{bd^3}{12}$

For circle $I = \frac{\pi d^4}{64}$

Polar 2nd moment of area for a circle $J = \frac{\pi d^4}{32}$

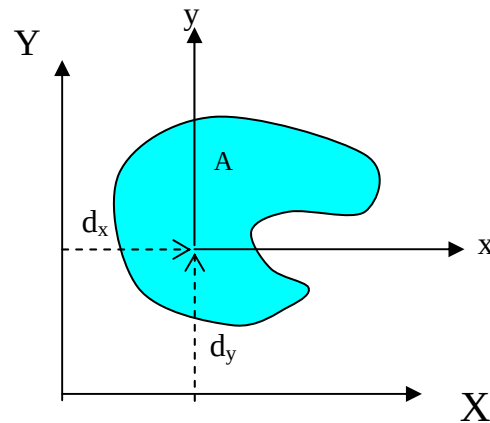
Parallel axis theorems

When centroidal axis is different from the segment axis then parallel axis theorem, as defined below, needs to be applied to find the I values for the overall section about axes passing through the centroid of the overall section.

$$I_{XX} = \sum (I_{xx} + Ad_y^2)$$

$$I_{YY} = \sum (I_{yy} + Ad_x^2)$$

$$I_{XY} = \sum (I_{xy} + Ad_x d_y)$$



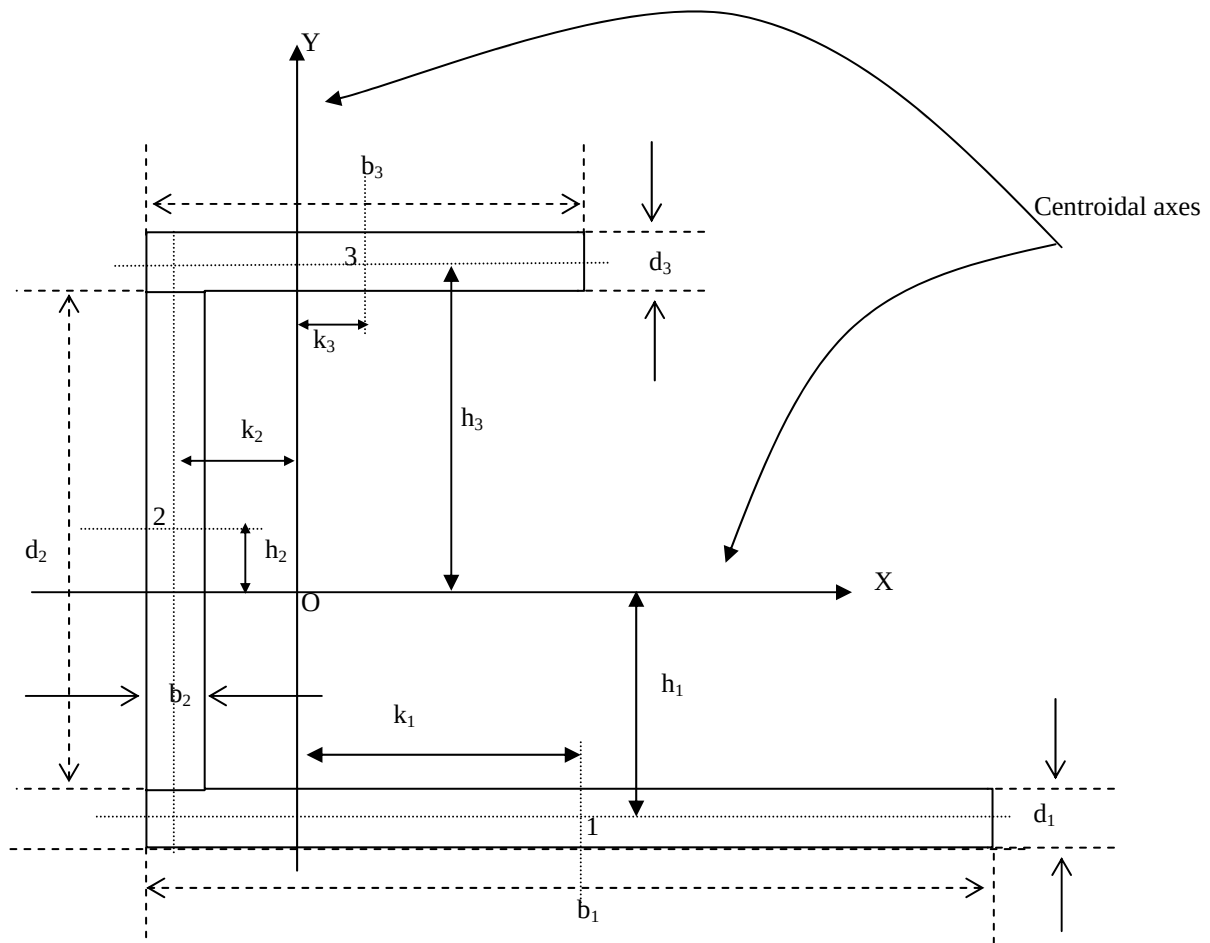
For idealised section above can be reduced to

$$I_{XX} = \sum (Ad_y^2)$$

$$I_{YY} = \sum (Ad_x^2)$$

$$I_{XY} = \sum (Ad_x d_y)$$

Application to the complex regular example showed previously.



$$I_{XX} = \sum (I_{xx} + A h_i^2) = \sum_{i=1-3} \left(\frac{b_i d_i^3}{12} + b_i d_i h_i^2 \right)$$

$$I_{YY} = \sum (I_{yy} + A k_i^2) = \sum_{i=1-3} \left(\frac{d_i b_i^3}{12} + b_i d_i k_i^2 \right)$$

Product of the moment of area;

Is defined as

$$I_{xy} = \int_A xy dA$$

Which is usually zero for regular symmetrical sections, as mentioned above.

When applied to a composite section the above equation it transforms into the Parallel axis theorem, which is represented as;

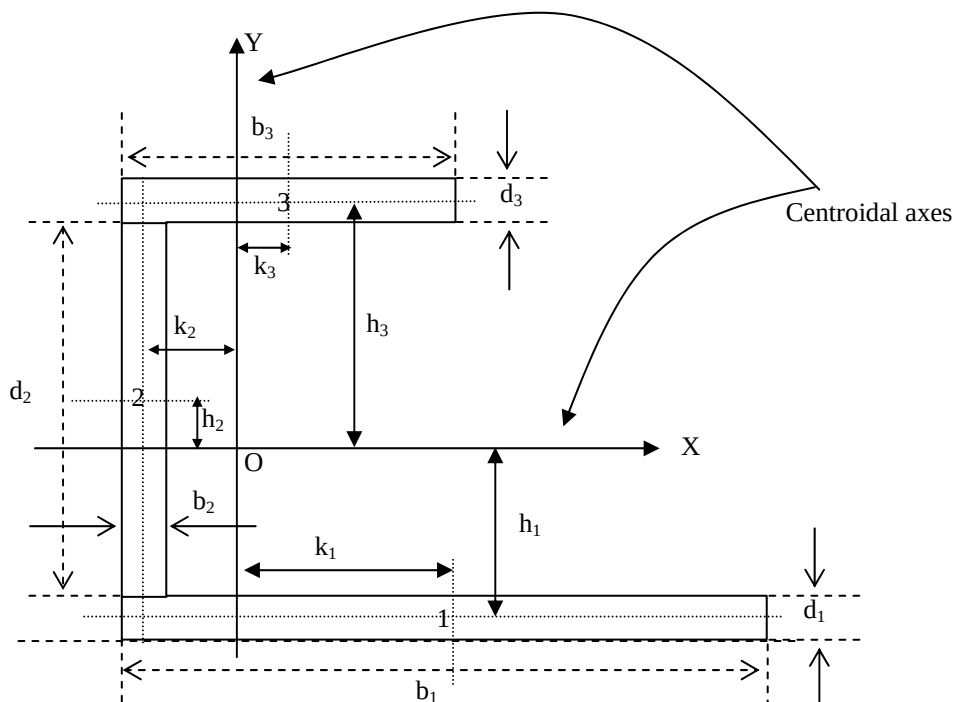
$$I_{XY} = \sum (I_{xy} + A d_x d_y)$$

The Composite section can be divided into a number of regular symmetrical segments. The I_{xy} about the local centroidal axis of each segment is zero. What is left is the parallel axis terms, which is;

$$I_{XY} = \sum (A d_x d_y)$$

Where d_x and d_y are the distances between the centroid of each segment and the centroid of the composite section.

The product of the moment of area for the example shown below is

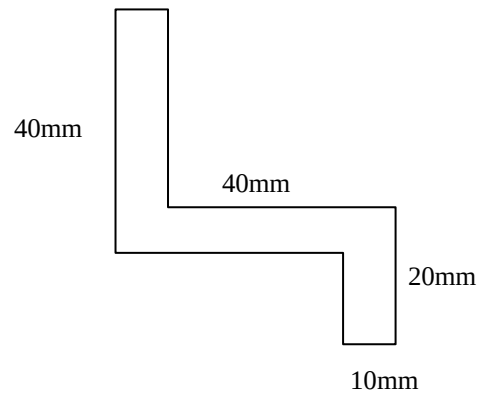
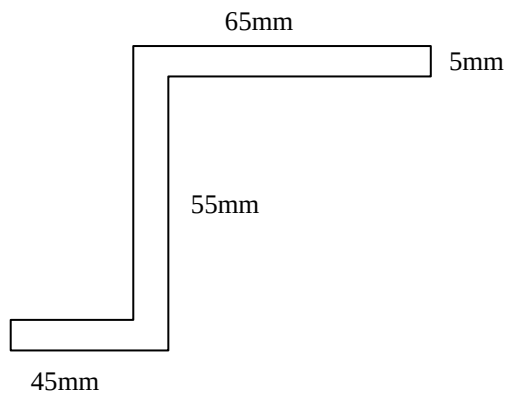
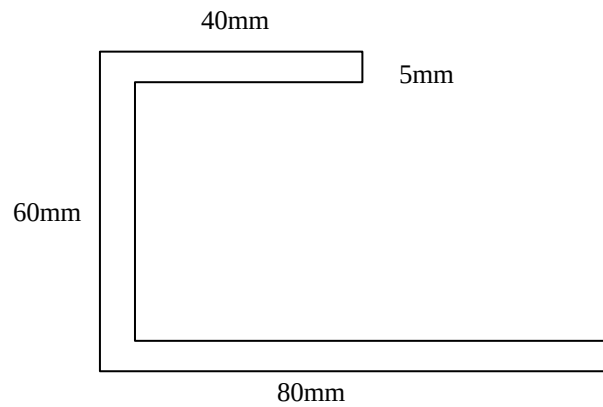
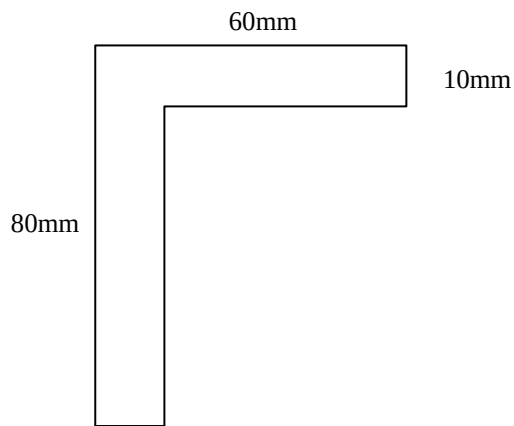


$$I_{xy} = \sum_{i=1-3} A_i d_{x_i} d_{y_i} = \sum_{i=1-3} b_i d_i h_i k_i$$

Here the values for h_i and k_i can be either positive or negative depending on the quadrant where the centroid of each segment lies.

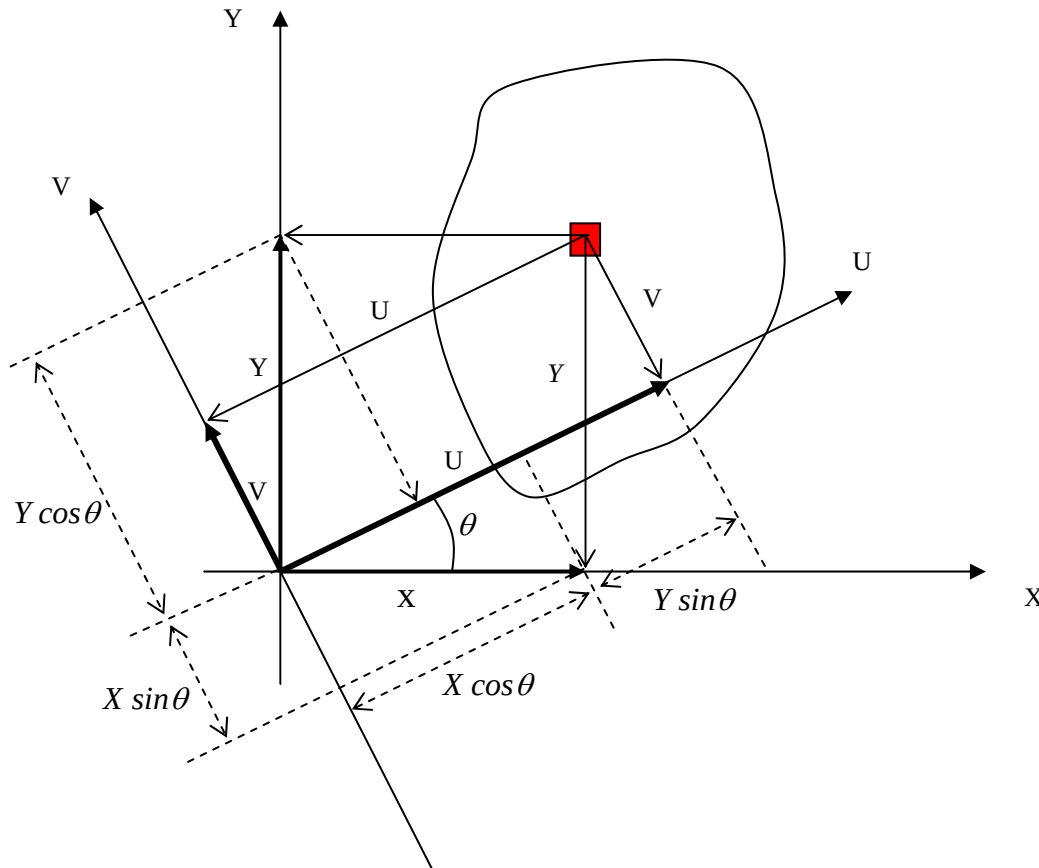
Tutorials:

Calculate I_{xx} and I_{yy} and I_{xy} for the following sections;



Unsymmetrical sections

If I_{xy} is not zero then X and Y axes will not be the principal axes so I_{xx} and I_{yy} are not the principal second moments of area.



An unsymmetrical section, shown above, has a Cartesian co-ordinates system axes X and Y and the Principal axes system U and V. The co-ordinate transformation from XY axes to UV axes takes the following form where

$$\begin{aligned} U &= X \cos \theta + Y \sin \theta \\ V &= Y \cos \theta - X \sin \theta \end{aligned} \quad (1)$$

Assuming that I_{xx} , I_{yy} and I_{xy} can be calculated using the classical methods, shown previously, we can use them to calculate I_{UU} and I_{VV} and I_{UV} about a new set of axes U and V: -

$$I_{UU} = \int_A V^2 dA$$

$$I_{VV} = \int_A U^2 dA$$

$$I_{UV} = \int_A UV dA$$

Substituting for U and V from (1) in the above equations results in:

$$\begin{aligned} I_{UU} &= I_{xx} \cos^2 \theta + I_{yy} \sin^2 \theta - 2I_{xy} \sin \theta \cos \theta \\ I_{VV} &= I_{xx} \sin^2 \theta + I_{yy} \cos^2 \theta + 2I_{xy} \sin \theta \cos \theta \\ I_{UV} &= I_{xx} \sin \theta \cos \theta - I_{yy} \sin \theta \cos \theta + I_{xy} (\cos^2 \theta - \sin^2 \theta) \end{aligned}$$

Or in terms of double angle:

$$\begin{aligned} I_{UU} &= \frac{(I_{xx} + I_{yy})}{2} + \frac{(I_{xx} - I_{yy})}{2} \cos 2\theta - I_{xy} \sin 2\theta \\ I_{VV} &= \frac{(I_{xx} + I_{yy})}{2} - \frac{(I_{xx} - I_{yy})}{2} \cos 2\theta + I_{xy} \sin 2\theta \\ I_{UV} &= \frac{1}{2} (I_{xx} - I_{yy}) \sin 2\theta + I_{xy} \cos 2\theta \end{aligned}$$

If X and Y are the principal axis then $I_{xy}=0$ and $I_{UU}=I_{xx}$ and $I_{VV}=I_{yy}$

If U and V are Principal axis then $I_{XY} \neq 0$ but $I_{UV} = 0$, the $I_{UU}=I_{\max}$ and $I_{VV}=I_{\min}$

Thus

$$\tan 2\theta = \frac{-2I_{xy}}{I_{xx} - I_{yy}}$$

Knowing the above equation we can find an equivalent expressions for $\sin 2\theta$ and $\cos 2\theta$. Substituting then in the expression for I_{UU} and I_{VV} , shown previously, results in the following two equations.

$$I_{Max} = \frac{1}{2}(I_{xx} + I_{yy}) + \frac{1}{2}\sqrt{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}$$

$$I_{Min} = \frac{1}{2}(I_{xx} + I_{yy}) - \frac{1}{2}\sqrt{(I_{xx} - I_{yy})^2 + 4I_{xy}^2}$$

Above expression can be used to calculate the principal second moment of an irregular area.

These are important when calculating;

- Stress due to bending of an unsymmetrical section.
- The buckling load of a column with an un-symmetrical cross-section.

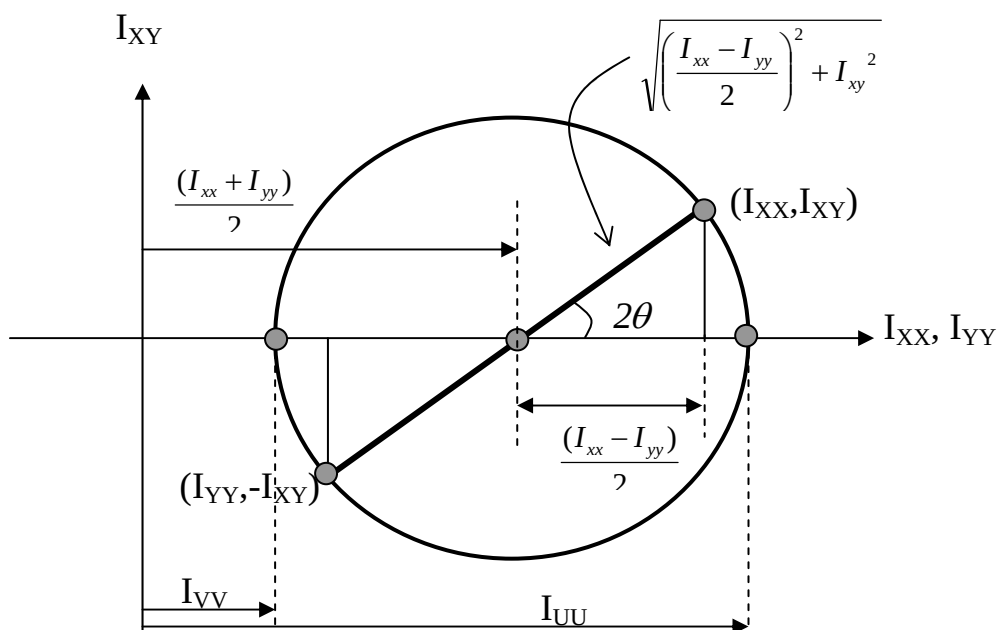
Graphical Method (Mohr's circle)

The equations for the second moment (I_{UU}) and the product moments (I_{UV}) of area can be squared and added to form the equation for a Mohr's circle for section property. This can be shown graphically using a Mohr's with centre at $\frac{I_x + I_y}{2}$ and

radius of $\sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2}$ on axes I_{xx} or I_{yy} and I_{xy} .

Method approach:-

1. At a point on a graph paper draw a horizontal axis representing I_{xx} & I_{yy} .
2. At a point on that axis draw a vertical axis I_{xy} .
3. Where these two axes meet is will be the origin O.
4. Select two points A and B with the following co-ordinates.
 $A(I_{xx}, I_{xy})$ and $B(I_{yy}, -I_{xy})$.
5. These points are on the appropriate ends of a diameter AB of the Mohr's circle.
6. A circle can now be drawn with centre C, where the straight line AB intersects the horizontal axis I_{xx} / I_{yy} , and radii CA or CB.
7. This circle will intersect the horizontal axis at points D and E.
8. $OD = I_{UU}$ the maximum principal second moment of area.
9. $OE = I_{VV}$ the minimum principal second moment of area for this unsymmetrical section.
10. Angle θ is the orientation between the local Cartesian co-ordinate axes systems and the principal axes of the section.
11. Note that Mohr's circle is in terms of Double angle 2θ



Summary of Important Notes:-

- i) $I_{xy} = 0$ if the composite section has at least one axes of symmetry
- ii) $I_{xy} \neq 0$ if there are no axis of symmetry.
- iii) Unlike I_x and I_y , which are always positive numbers, I_{xy} can be either a positive or a negative number depending on shape of the composite section.
- iv) When composite section has at least one axis of symmetry, then X and Y are called the principal axes and I_{xx} and I_{yy} are called the principal (Max and Min) second moments of area for that section.
- v) If section has no axis of symmetry, then $I_{xy} \neq 0$, X & Y axes are not the principal axes and I_{xx} and I_{yy} are not the Principal (Max or Min) second moments of area for that section.
- vi) For completely unsymmetrical section there exists another set of axes I_U and I_V , which are called principal axes.
- vii) The magnitude and orientation of these values will be different from those about the X and Y axes.
- viii) They can be calculated using either graphical or analytical methods as will be discussed later.

Approximation for thin walled sections

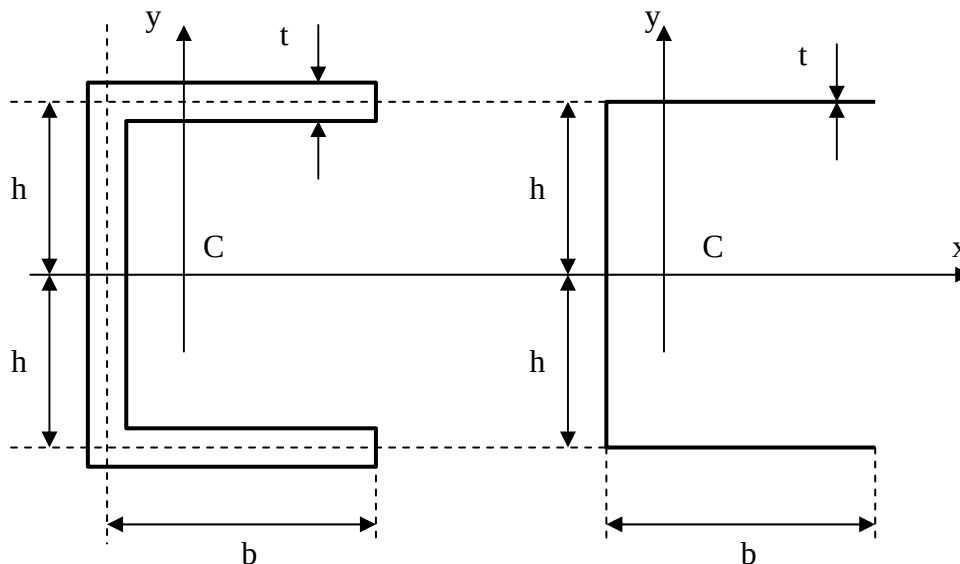
The stiffeners used in aircraft structures have thin walled sections. The analysis of such components requires some simplifying assumptions to be made when determining their section properties. Some of these assumptions are listed below:

- Thickness t is small compared other cross section dimensions.
- Neglect square and higher powers of t in the calculation of section properties
- Sections are represented by the mid-line of its wall.

This is best be demonstrated with the help of the following examples:

Example 1

Derive an expression for I_{xx} for channel section shown below.

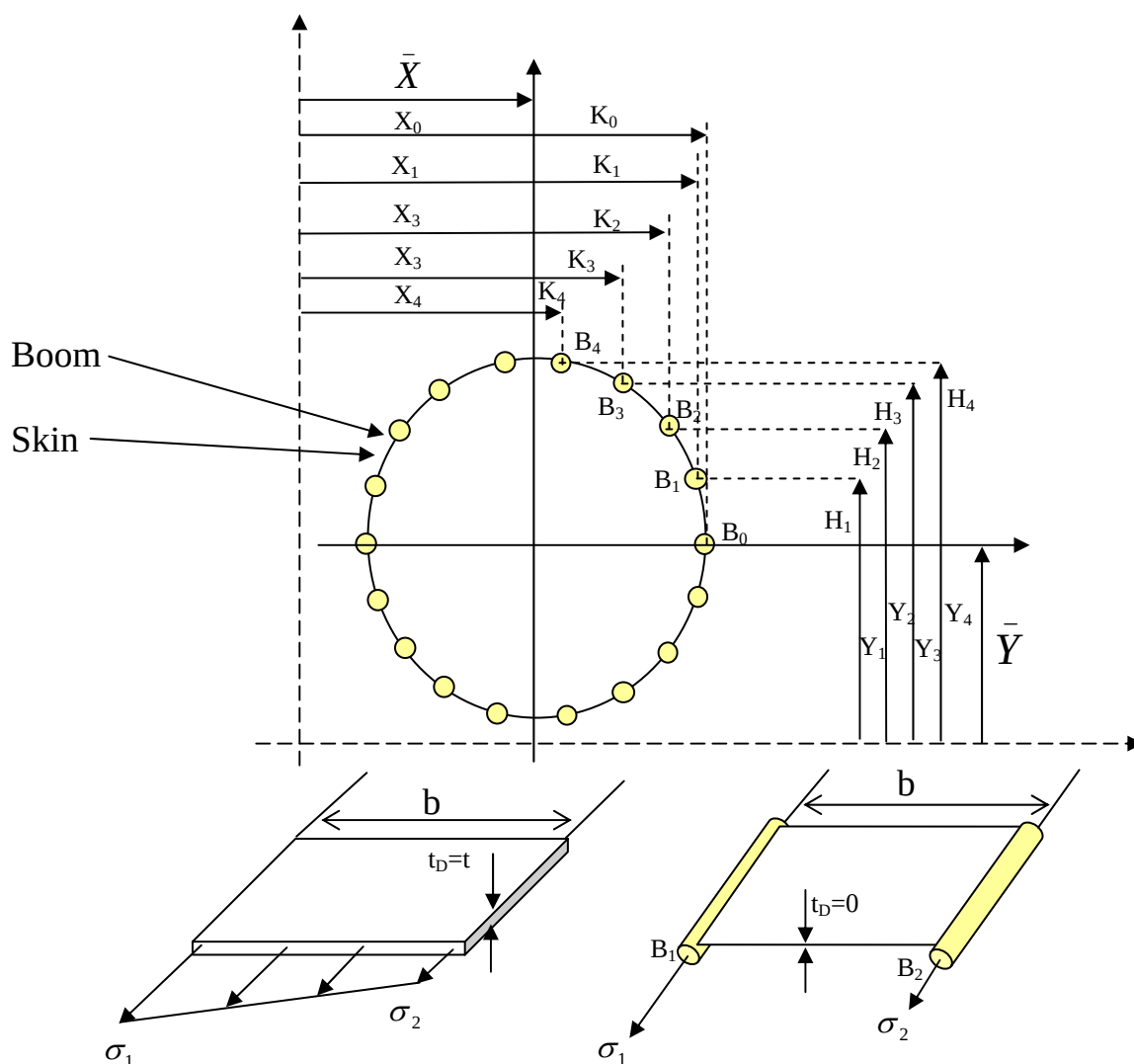


Ignoring the square and higher order t_s , above equation reduces to;

$$I_{xx} = 2bth^2 + t \frac{(2h)^3}{12}$$

Application to aircraft structures

What we have discussed so far is valid, but only directly applies to simple sections. However, aircraft structures are more complicated and a certain degree of simplification is necessary in order to handle the vast amount of calculation involved. For aircraft structures it would be reasonable to assume that the direct stresses are constant on stringer's cross section and are carried by them. Thus we can replace the stringers and spar flanges by concentrations of area known as BOOMS, over which the direct stress is constant. The booms are located along the mid line of the skin. Another valid assumption is that the skin in between only experiences shear.



Idealisation

The panel on the left is shown in its actual form. On the right in an idealised form where the panel thickness $t_D=0$. The linear stress distribution on the left is replaced by extreme values acting on the booms. Since the stress in the actual and idealised panel must be the same we can obtain an expression for an equivalent area for the booms B_1 and B_2 . Taking moments about the R H edge of each panel; we get:

$$\sigma_2 t_D \frac{b^2}{2} + \frac{1}{2}(\sigma_1 - \sigma_2) t_D b \frac{2}{3} b = \sigma_1 B_1 b$$

From which

$$B_1 = \frac{t_D b}{6} \left(2 + \frac{\sigma_2}{\sigma_1} \right)$$

Similarly

$$B_2 = \frac{t_D b}{6} \left(2 + \frac{\sigma_1}{\sigma_2} \right)$$

For axial load only $\frac{\sigma_1}{\sigma_2} = 1$ and $B_1=B_2=t_D b/2$, for pure bending $\frac{\sigma_1}{\sigma_2} = -1$ and $B_1=B_2 = t_D b/6$. Thus different loading requires different idealisations.

The centroid of such idealised section can be calculate using the following relation ships.

$$\bar{Y} = \frac{\sum B_i Y_i}{\sum B_i}$$

And

$$\bar{X} = \frac{\sum B_i X_i}{\sum B_i}$$

The moment of inertia about the axes can be idealised as

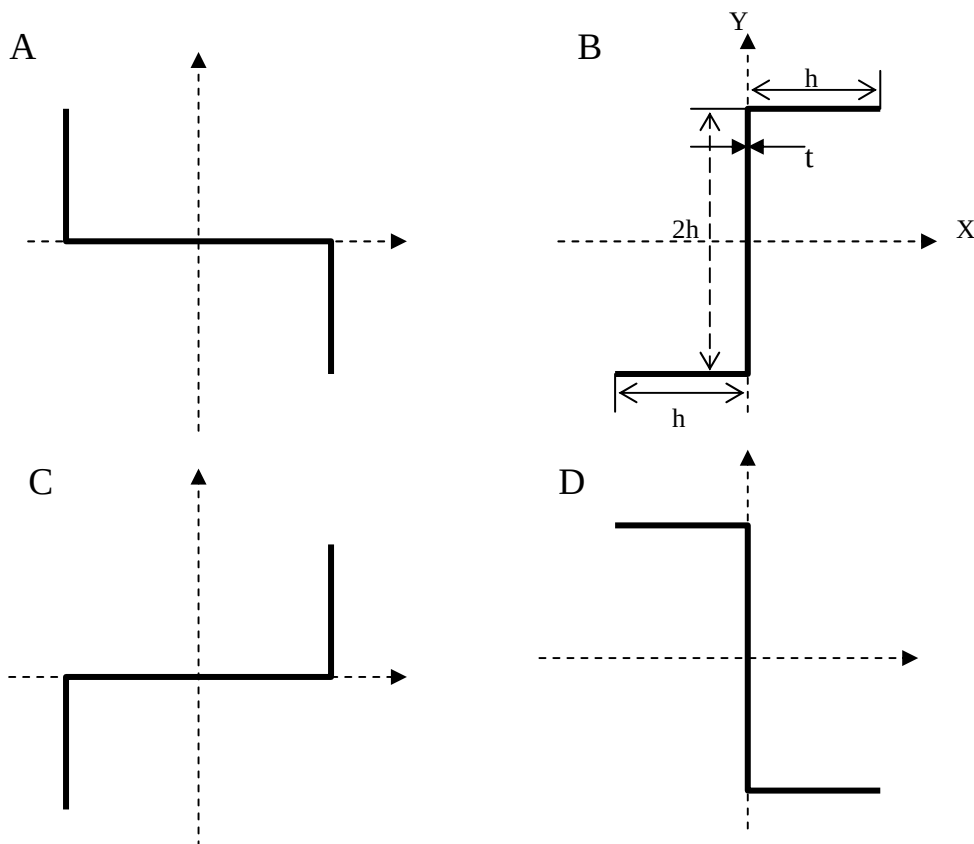
$$I_{xx} = \sum B_i h_i^2$$

$$I_{yy} = \sum B_i K_i^2$$

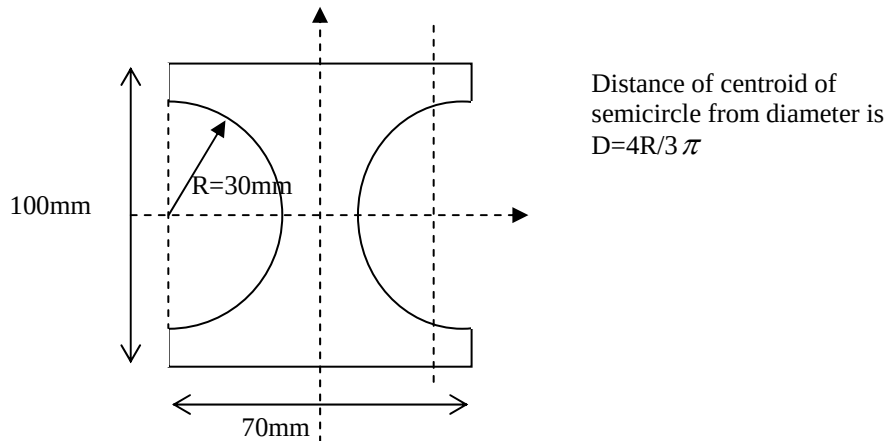
$$I_{xy} = \sum B_i K_i h_i$$

Tutorials

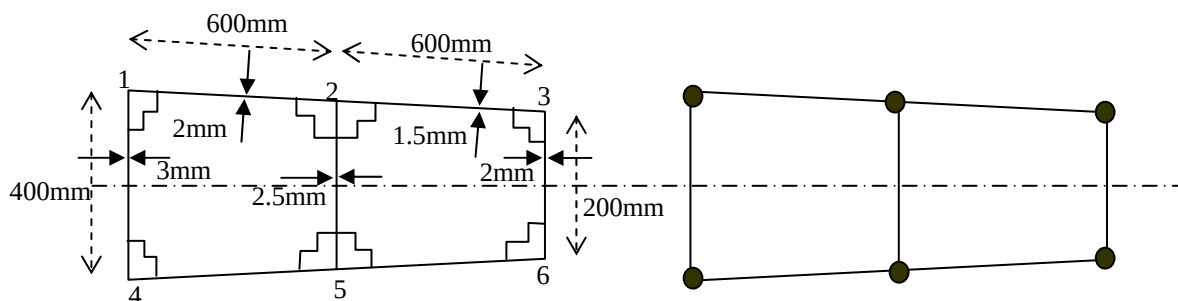
1) Calculate, in terms of **h** and **t**, both the exact and the approximate expressions for I_{xx} , I_{yy} , I_{xy} , I_{UU} and I_{VV} for the thin walled sections shown below. If $h = 100\text{mm}$ and $t = 1\text{mm}$, verify the values of I_{UU} , I_{VV} and their orientation using graphical method (Mohr's circle for I).



2) Find I_{xx} and I_{yy} for the section shown below

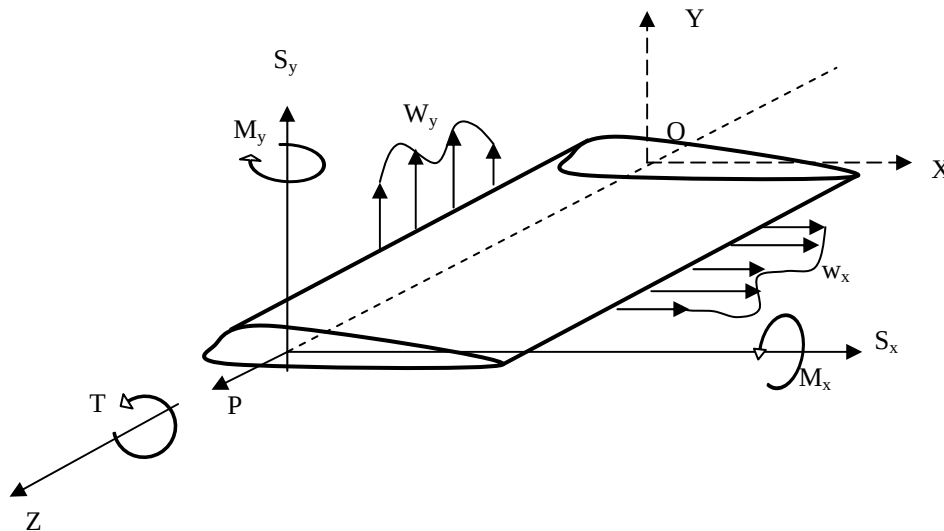


3) Part of a wing section is in the form of the two cell-box shown below in which the vertical spars are connected to the wing skin through angle section all having cross section area of 300mm^2 . Idealise this section into an arrangement of direct stress carrying booms and shear panels suitable for resisting bending moment in vertical plane. Position the booms at the spar/skin junctions. Also calculate I_{xx} and I_{yy} for this section.



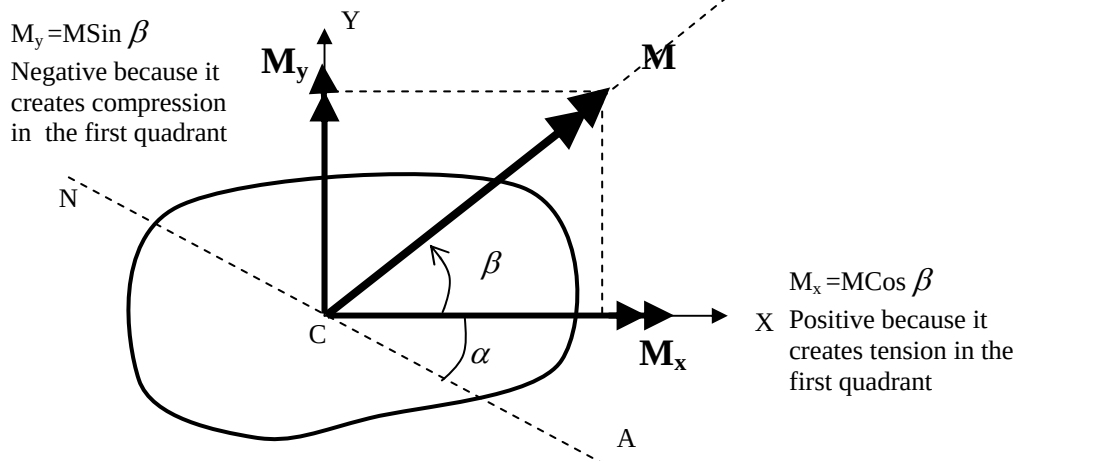
Un-symmetrical Bending of beams

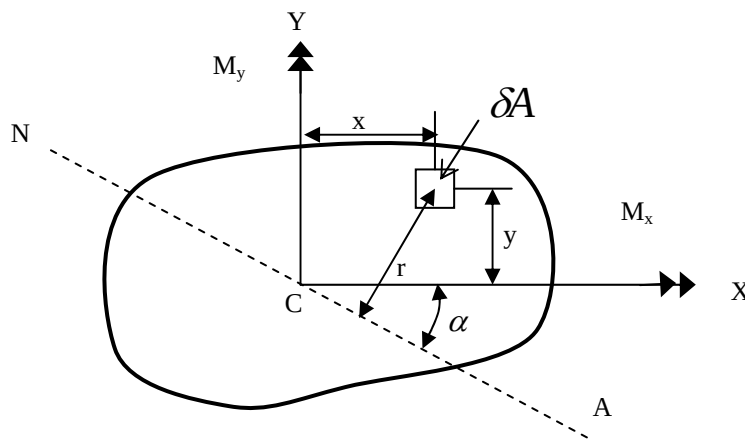
Here, it is of no consequence whether or not the section is open or closed. The general theory of bending will be developed based on assumptions that the material is homogeneous and linearly elastic with plane sections remaining plane.



- The procedure starts with deciding sign convention and symbols.
 - M_x and M_y are positive when they introduce tension in the first quadrant.
 - M_x is BM about X axis
 - M_y is BM about Y axis
 - S_x is shear force along X axis
 - S_y is shear force along y axis
 - W is load intensity associated with distributed loading
 - P is direct external axial load
 - T is the torque

- Direct stress due to bending.





Consider a beam having the section shown above, supporting bending moments M_x and M_y and bending about a neutral axis which passes through the centroid C. The direct stress σ_z on an element δA at a point (x, y) at a distance r from the neutral axis is given by

$$\sigma_z = E \varepsilon_z$$

If the beam is bent to a radius of curvature R about the NA, then:

$$\varepsilon_z = \frac{r}{R}$$

Substituting this in the above equation we get:

$$\sigma_z = E \frac{r}{R}$$

Since the beam supports pure bending the resultant normal force must be in state of equilibrium.

$$\therefore \int_A \sigma_z dA = 0$$

Replacing σ_z in this equation with its equivalent from above and ignoring constants.

$$\int_A r dA = 0$$

This means the first moment of area about NA is zero. Therefore the NA passes through centroid as stated above. Suppose the NA makes an angle α measured CCW relative to Cx, then

$$r = x \sin \alpha + y \cos \alpha$$

And from above:

$$\sigma_z = \frac{E}{R}(x \sin \alpha + y \cos \alpha) \quad (A)$$

The internal moments have the same sense as the applied Bending Moments.

$$\therefore M_x = \int_A \sigma_z y dA,$$

&

$$-M_y = \int_A \sigma_z x dA$$

Substitute for σ_z from equation (A) into above equation and defining the second moments of area about Cx and Cy as the followings

$$I_{xx} = \int_A y^2 dA$$

$$I_{yy} = \int_A x^2 dA$$

$$I_{xy} = \int_A xy dA$$

Results in:

$$M_x = \frac{E \sin \alpha}{R} I_{xy} + \frac{E \cos \alpha}{R} I_{xx}$$

$$-M_y = \frac{E \sin \alpha}{R} I_{yy} + \frac{E \cos \alpha}{R} I_{xy}$$

Or in matrix form:

$$\begin{Bmatrix} M_x \\ -M_y \end{Bmatrix} = \frac{E}{R} \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix}$$

From which:

$$\frac{E}{R} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \begin{bmatrix} I_{xy} & I_{xx} \\ I_{yy} & I_{xy} \end{bmatrix}^{-1} \begin{Bmatrix} M_x \\ -M_y \end{Bmatrix}$$

$$\frac{E}{R} \begin{Bmatrix} \sin \alpha \\ \cos \alpha \end{Bmatrix} = \frac{1}{I_{xx}I_{yy} - I_{xy}^2} \begin{bmatrix} -I_{xy} & I_{xx} \\ I_{yy} & -I_{xy} \end{bmatrix} \begin{Bmatrix} M_x \\ -M_y \end{Bmatrix}$$

So from equation (A)

$$\sigma_z = \left(\frac{-M_y I_{xx} - M_x I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} + M_y I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) y$$

or

$$\sigma_z = - \left(\frac{M_y I_{xx} + M_x I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} + M_y I_{xy}}{I_{xx}I_{yy} - I_{xy}^2} \right) y$$

Or re-arranged into:

$$\sigma_z = \frac{M_x (I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} - \frac{M_y (I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2}$$

Additional rules to comply with the variations in textbooks.

In the event of using positive M_x and M_y for the development of this equation, The above equations take a slightly different forms as shown below. In this case we need to use a sign convention based on the Right Hand Rule (RHR) to determine the accurate signs for the bending moments.

$$\sigma_z = \left(\frac{M_y I_{xx} - M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) x + \left(\frac{M_x I_{yy} - M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2} \right) y$$

Or rearranged into:

$$\sigma_z = \frac{M_x (I_{yy} y - I_{xy} x)}{I_{xx} I_{yy} - I_{xy}^2} + \frac{M_y (I_{xx} x - I_{xy} y)}{I_{xx} I_{yy} - I_{xy}^2}$$

The right hand rule implies that:

- 1) We use absolute values for M_x and M_y
- 2) Use RH cork-screw rule to decide the effect of these moment on the first quadrant.
- 3) If M generates **Tension** in the **1st quadrant**, the sign for absolute M is Positive
- 4) If M generates **Compression** in the **1st quadrant** the sign for absolute M is Negative

We can see if for example that is $M_y=0$, M_x produces a stress which is a function of both x and y and vice versa. For cases where either (or both) C_x or C_y are axis of symmetry the $I_{xy}=0$ and above equation reduces to:

$$\sigma_z = \frac{M_x y}{I_{xx}} - \frac{M_y x}{I_{yy}}$$

If either M_x or M_y is zero then

$$\sigma_z = \frac{M_x y}{I_{xx}} \quad \text{Or} \quad \sigma_z = \frac{-M_y x}{I_{yy}}$$

Position of Neutral Axis

At all points of the neutral axis the direct stress $\sigma_z = 0$, therefore

$$0 = -\left(\frac{M_y I_{xx} + M_x I_{xy}}{I_{xx} I_{yy} - I_{xy}^2}\right)x + \left(\frac{M_x I_{yy} + M_y I_{xy}}{I_{xx} I_{yy} - I_{xy}^2}\right)y \quad (B)$$

$$\frac{y}{x} = \tan \alpha = \frac{M_y I_{xx} + M_x I_{xy}}{M_x I_{yy} + M_y I_{xy}} = \text{slope}$$

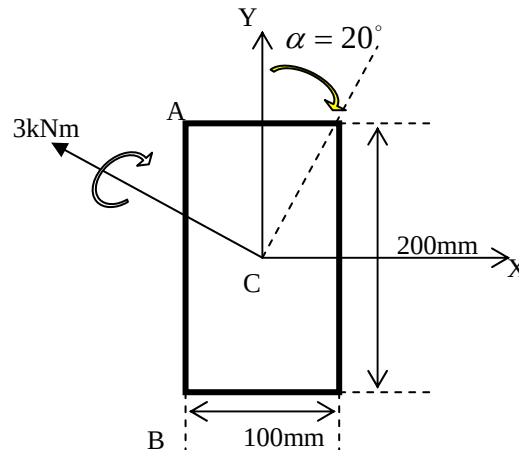
$y=mx$ where m is calculated from the above equation.

All you have to do now is to draw the straight-line $y=mx$ through the origin/centroid.

Example 1

The Couple M is applied to the following beam in the plane forming an angle α with the vertical. Determine

1. Stress at A
2. Stress at B
3. The angle that NA makes with the horizontal axis



$$\sigma_z = \frac{M_x(I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} - \frac{M_y(I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2}$$

Apply the above equation to this problem. Due to symmetry, the above equation reduces to . Using the RH Rule:

$$\sigma_z = \frac{M_x y}{I_{xx}} - \frac{M_y x}{I_{yy}}$$

where

$$M_x = M \cos \alpha = 3 \times 10^6 \cos 20^\circ \text{ Nmm} \quad (\text{It's effect on 1}^{\text{st}} \text{ quadrant is a compressive (-)})$$

$$M_y = M \sin \alpha = 3 \times 10^6 \sin 20^\circ \text{ Nmm} \quad (\text{It's effect on 1}^{\text{st}} \text{ quadrant is a compressive (-)})$$

$$I_x = \frac{bd^3}{12} = \frac{100 \times 200^3}{12} \text{ and } I_y = \frac{db^3}{12} = \frac{200 \times 100^3}{12}$$

$$\sigma_A = -\frac{M_x(100)}{I_{xx}} - \frac{M_y(-50)}{I_{yy}}$$

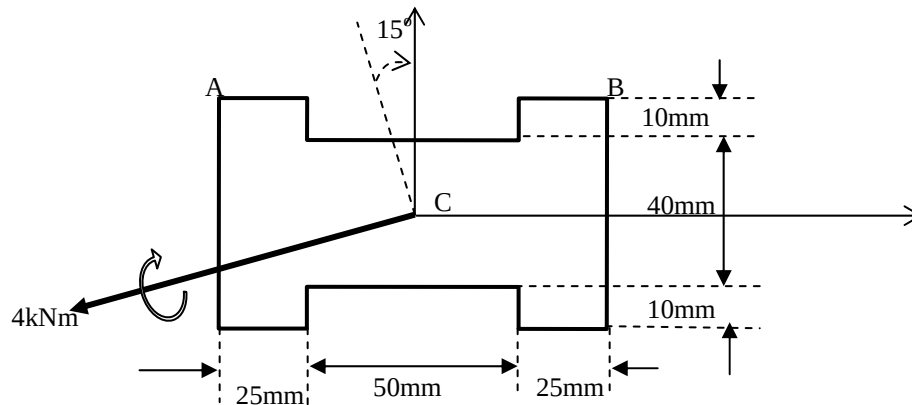
$$\sigma_B = -\frac{M_x(-100)}{I_{xx}} - \frac{M_y(-50)}{I_{yy}}$$

Neutral axis is given by $\tan \phi = \left(\frac{M_y I_x}{M_x I_y} \right) = m$ or $Y = m X$

Example 2

The Couple M is applied to the following beam in the plane forming an angle α with the vertical. Determine

1. Stress at A
2. Stress at B
3. The angle that NA makes with the horizontal axis



We start with

$$\sigma_z = \frac{M_x(I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} - \frac{M_y(I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2}$$

Apply the above equation to this problem. Due to symmetry, the above equation reduces to . Using the RH Rule:

$$\sigma_z = \frac{M_x y}{I_{xx}} - \frac{M_y x}{I_{yy}}$$

where

$$M_x = M \cos \alpha = 4 \times 10^6 \cos 195 \text{ Nmm} \quad (\text{It's effect on 1}^{\text{st}} \text{ quadrant is Compressive (-)})$$

$$M_y = M \sin \alpha = 4 \times 10^6 \sin 195 \text{ Nmm} \quad (\text{It's effect on 1}^{\text{st}} \text{ quadrant is Tensile (+)})$$

$$I_x = \sum \frac{bd^3}{12} = 2\left(\frac{25 \times 60^3}{12}\right) + \frac{50 \times 40^3}{12}$$

and

$$I_y = \sum \frac{db^3}{12} + Ah^2 = \left(\frac{40 \times 50^3}{12}\right) + 2\left(\left(\frac{60 \times 25^3}{12}\right) + (25 \times 60) \times 37.5^2\right)$$

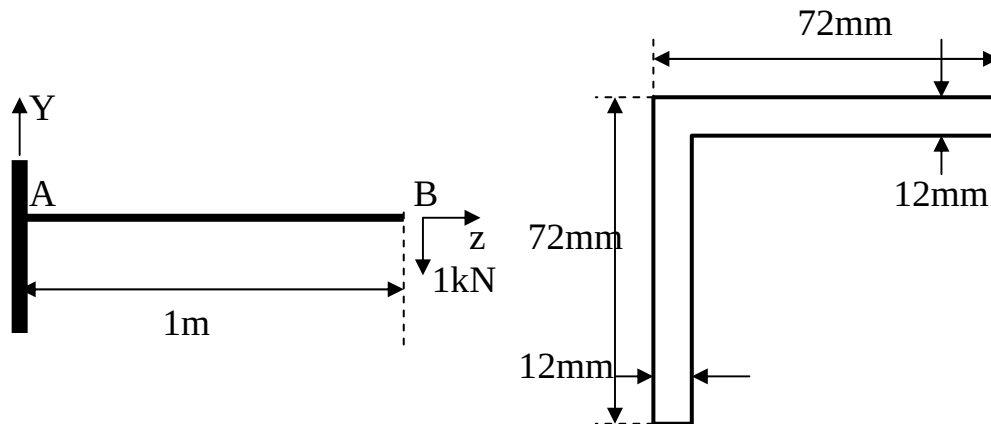
$$\sigma_A = \frac{M_x(+30)}{I_{xx}} - \frac{M_y(-50)}{I_{yy}}$$

$$\sigma_B = \frac{M_x(+30)}{I_{xx}} - \frac{M_y(+50)}{I_{yy}}$$

Neutral axis is given by $\tan \phi = \left(+\frac{M_y I_x}{M_x I_y}\right) = m \quad \text{or} \quad Y = m X$

Examples 3

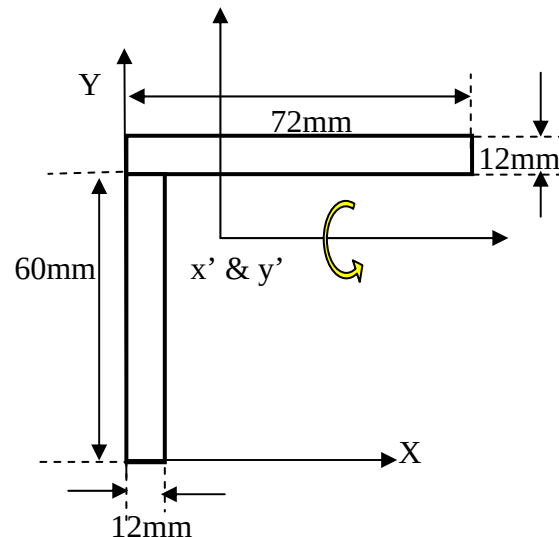
Determine the maximum bending stress in a cantilever of length 1m, loaded as shown. The section is illustrated below.



Solution

Maximum bending moment is at the built in end.

Centre of area (Centroid).



$$\bar{Y} = \frac{\sum y \delta A}{\sum \delta A}$$

$$\bar{Y} = \frac{(72 \times 12)(72 - 6) + (60 \times 12)(30)}{72 \times 12 + 60 \times 12} = 49.64 \text{ mm}$$

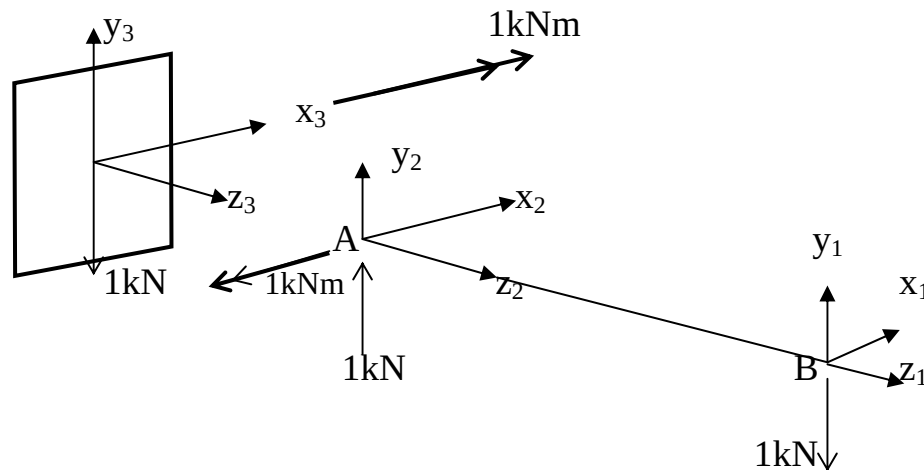
$$\bar{X} = \frac{\sum x \delta A}{\sum \delta A}$$

$$\bar{X} = \frac{(72 \times 12)(36) + (60 \times 12)(6)}{72 \times 12 + 60 \times 12} = 22.36 \text{ mm}$$

$$I_{xx} = \frac{72 \times 12^3}{12} + 72 \times 12 \times (66 - 49.64)^2 + \frac{12 \times 60^3}{12} + 60 \times 12 \times (49.64 - 30)^2 = 735343 \text{ mm}^4$$

$$I_{yy} = \frac{12 \times 72^3}{12} + 72 \times 12 \times (36 - 22.36)^2 + \frac{60 \times 12^3}{12} + 60 \times 12 \times (22.36 - 6)^2 = 735343 \text{ mm}^4$$

$$I_{xy} = (72 \times 12)(66 - 49.64)(36 - 22.36) + (60 \times 12)(6 - 22.36)(30 - 49.64) = 424145 \text{ mm}^4$$



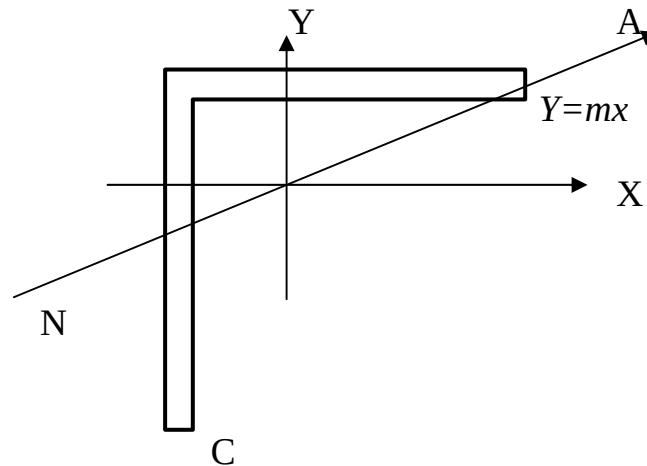
$M_x = 1 \text{ kNm}$ or $1 \times 10^6 \text{ Nmm}$ and $M_y = 0$ and $\sigma_{zz} = 0$ on the neutral axis.

Hence from (B) and using Right Hand Rule '(RHR)' M_x is Positive 1kNm or 1000Nm and $M_y = 0$,

$$1 \times 10^6 (735343 \times Y - 424145 \times X) - 0 \times (735343 \times X - 424145 \times Y) = 0$$

Therefore $735343 Y = 424145 X$, Hence $Y/X = 0.577$ Hence $\alpha = 30^\circ$

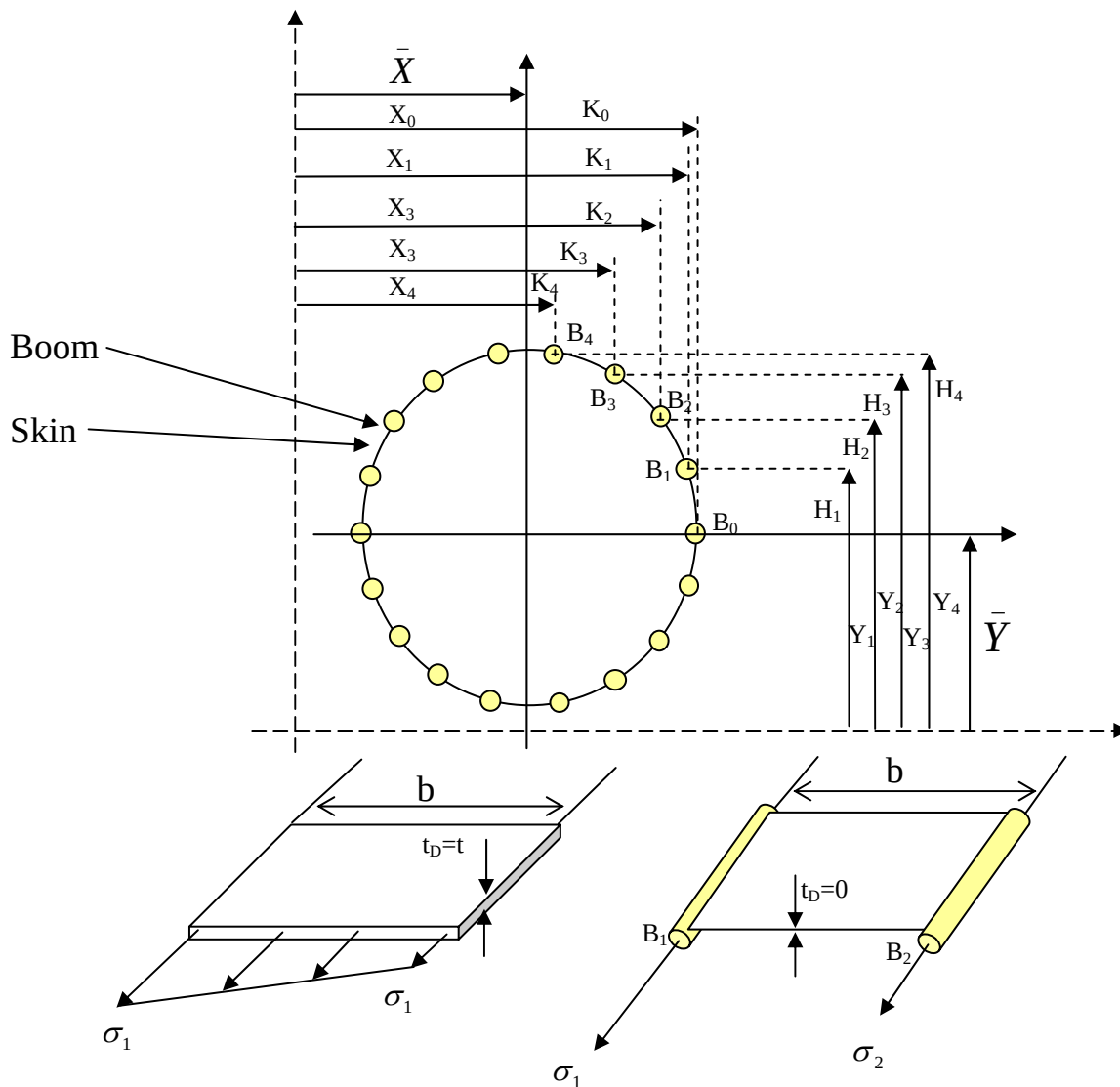
By drawing a scale diagram of the section one can see that point C is the furthest from the neutral axis, and will therefore have the greatest stress.



At C, $X = -10.36 \text{ mm}$; $Y = -49.64 \text{ mm}$

$$\sigma_{zz} = \frac{1000(0.735 \times 10^{-6} \times (-0.04964) - 0.424 \times 10^{-6} \times (-0.01036))}{(0.735 \times 0.735 - 0.424^2) \times 10^{-12}} = -89.15 \text{ MN/m}^2$$

Effects of idealisation on Bending



$$\sigma_z = \frac{M_x(I_{yy}y - I_{xy}x)}{I_{xx}I_{yy} - I_{xy}^2} - \frac{M_y(I_{xx}x - I_{xy}y)}{I_{xx}I_{yy} - I_{xy}^2}$$

The theory presented for simple bending and bending of unsymmetrical beams, shown above, also applies to idealised section shown above. The solution depends on the section being investigated. In these equations the co-ordinates (x,y) of a point on the section are referred to axes having their origin at the centroid of the direct stress carrying areas (BOOMS). I_{xx} , I_{yy} and I_{xy} are calculated for the direct stress carrying areas only. The method used here is the parallel axis theorem and in the forms shown below:

$$I_x = \sum B_i H_i^2$$

$$I_y = \sum B_i K_i^2$$

$$I_{xy} = \sum B_i H_i K_i$$

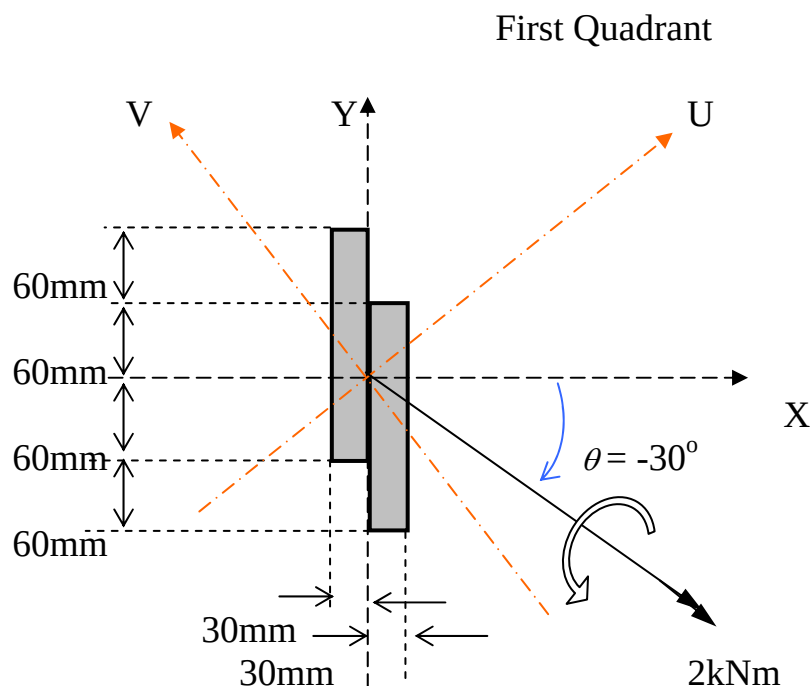
When a section is fully idealised the direct stress distribution will consist of a series of direct stresses concentrated at the centroid of each boom.

Tutorials

- 1) A cantilever beam is 2m long and is subjected to a moment of $M = 2 \text{ kNm}$ at its free end. The beam cross section is formed by nailing together two 30mm by 180mm boards, as shown below. The axis at which the moment acts makes an angle of -30° to the horizontal X-axis as shown.

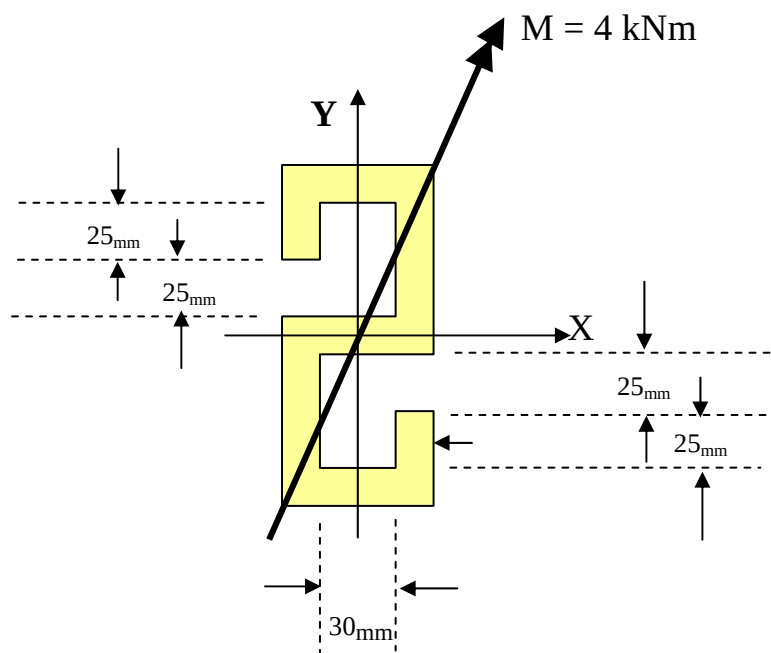
Determine;

- a) I_{xx} , I_{yy} and I_{xy} for the beam.
- b) The angle that the neutral axis forms with the horizontal plane.
- c) The position and the magnitude of the maximum tensile stress in the beam.



Positive Z is out of the plane of the paper.

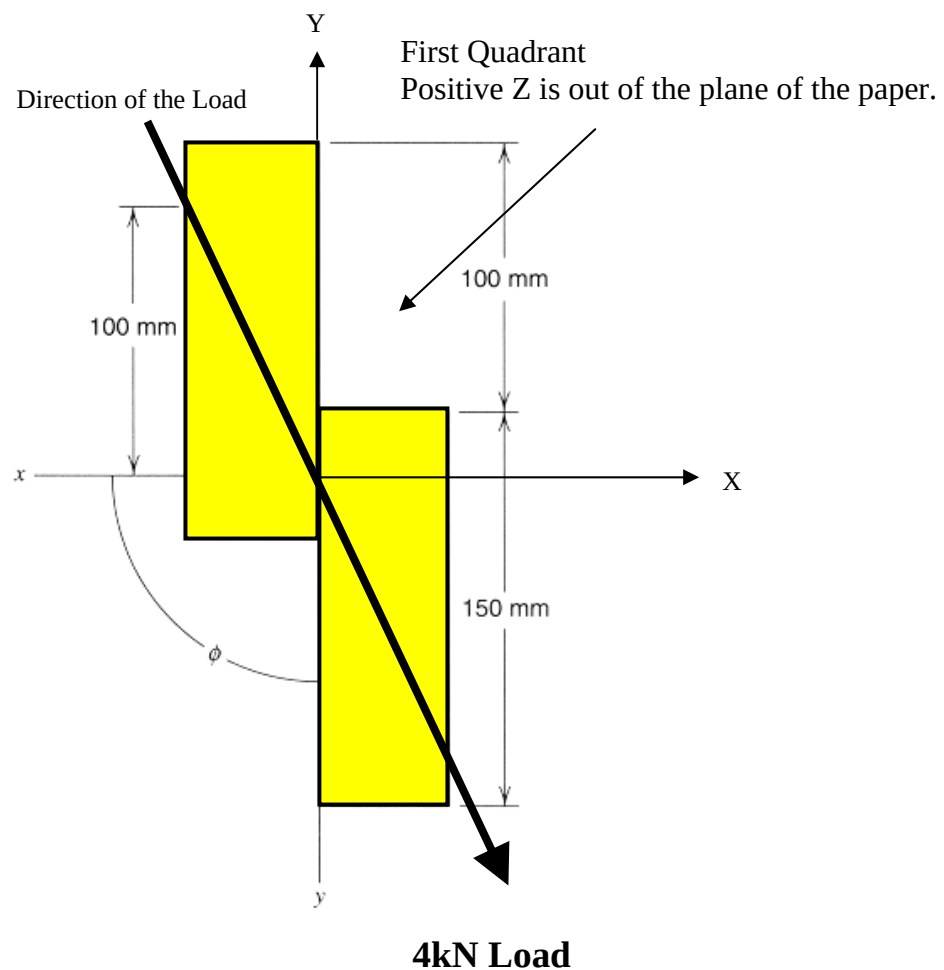
- 2) A simple cantilever that is 2 m long, with cross section shown below is subjected to a bending moment 4 kNm at its free end. The plane of the moment passes through the centroid of the section as shown. If the section has a uniform thickness of 15mm, $I_{xx} = 11.2\text{E}^{-6}\text{m}^4$ and $I_{yy} = 1.99\text{E}^{-6}\text{m}^4$, determine;
- I_{xy} for the beam.
 - The angle that the neutral axis forms with the horizontal plane.
 - The position and the magnitude of the maximum tensile stress in the beam.



- 3) A simple composite beam is subjected to a concentrated load $P = 4 \text{ kN}$ at mid length of a span of 2m. The beam cross section is formed by bonding together two 50mm by 150mm composite beams, as shown below. The plane of the load passes through the centroid of the two boards as indicated.

Determine;

- I_{xx} , I_{yy} and I_{xy} for the beam.
- The angle that the neutral axis forms with the horizontal plane.
- The position and the magnitude of the maximum tensile stress in the beam.



- 4) A simple wing spar is modelled as a cantilever beam of the cross section shown below is subjected to a bending moment of 25 kNm at its free end. The plane of the moment makes an angle of 15° with the vertical. The plane of the moment passes through the centroid of the section as shown. Determine;
- I_{XX} , I_{YY} and I_{XY} for the beam.
 - The angle that the neutral axis forms with the horizontal plane.
 - The position and the magnitude of the maximum tensile stress in the beam.

